

Protocol for the diagnostic performance of C reactive protein, procalcitonin and interleukin-6 for serious bacterial infections among children ≤36 months old presenting with fever without source: a systematic review and meta-analysis

Natalia Sutiman ¹, Sarah Hui Wen Yao,² Sharon Si Min Goh,² Rehena Sultana,³ Shu-Ling Chong ^{1,4}

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¹Department of Emergency Medicine, Kandang Kerbau Women's and Children's Hospital, Singapore

²Kandang Kerbau Women's and Children's Hospital, Singapore

³Center of Quantitative Medicine, SingHealth Duke-NUS Academic Medical Centre, Singapore

⁴Pediatric Academic Clinical Programme, Emergency Medicine Academic Clinical Programme, SingHealth Duke-NUS, Singapore

Correspondence to

Dr Natalia Sutiman; natalia.sutiman@mohh.com.sg

ABSTRACT

Introduction The management of fever without source in children ≤36 months old remains a diagnostic challenge as the underlying aetiologies can vary from self-limiting viral infections to serious bacterial infections (SBIs). Biomarkers such as C reactive protein (CRP), procalcitonin (PCT) and interleukin-6 (IL-6) have varying thresholds in the prediction of SBIs due to differences in SBI definitions, SBI prevalence, patient characteristics and timing of presentation. This protocol describes a systematic review and meta-analysis that aims to determine the thresholds at which CRP, PCT and IL-6 can perform optimally in distinguishing the presence of SBIs in children ≤36 months old, as well as to determine their performances in early detection of bacterial infections within 48 hours of fever onset.

Methods and analysis We will systematically search electronic databases including MEDLINE, Cochrane Central Register of Controlled Trials, Cochrane CENTRAL, EMBASE, CINAHL (Cumulative Index to Nursing and Allied Health Literature) and Science Citation Index from 1 July 2023 to 31 July 2023. We will include studies that report the diagnostic accuracy of CRP, PCT and IL-6 in detecting SBIs in children aged ≤36 months presenting with fever without apparent source. Randomised controlled trials (RCTs) and non-randomised studies including non-RCTs and controlled before-and-after studies will be included. A meta-analysis will be performed and diagnostic performances of these biomarkers will be reported.

Ethics and dissemination The results of this study will provide guidance on clinical decision-making in young children presenting with fever without source. Ethics approval will not be required for this study. The authors aim to publish the findings in a peer-reviewed journal as well as present at international conferences.

PROSPERO registration number CRD42023439093.

INTRODUCTION

Rationale

Fever without source, defined as fever less than 7 days without a clear source of infection,

WHAT IS ALREADY KNOWN ON THIS TOPIC

- ⇒ Management of fever without source in young children remains unclear.
- ⇒ Biomarkers such as C reactive protein (CRP), procalcitonin (PCT) and interleukin-6 (IL-6) have been well reported to accurately predict serious bacterial infections (SBIs).
- ⇒ However, optimal diagnostic thresholds and utility for early detection are debated.

WHAT THIS STUDY ADDS

- ⇒ This protocol describes a systematic review and meta-analysis that aims to determine the thresholds at which biomarkers such as CRP, PCT and IL-6 can accurately predict the presence of SBIs in children ≤36 months old, and to determine their diagnostic performances in early detection of bacterial infections within 48 hours of fever onset.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

- ⇒ The findings of this study will provide guidance on clinical decision-making in young children presenting with fever without source.

is one of the most common complaints in paediatric patients, representing approximately 20% of febrile children presenting to the emergency department.¹ It remains a diagnostic challenge as the underlying aetiologies can vary, from self-limiting viral infections to serious bacterial infections (SBIs) with long-term sequelae and mortality.² It is particularly challenging in children ≤36 months of age as they often present without focal symptoms and may even appear well.³ Despite extensive history and physical examination, clinicians are often unable to identify a specific source in this population.⁴ The well-appearing child

may still have an SBI including occult bacteraemia. The risk of missed diagnosis often leads to inappropriate prescribing of antibiotics, hospital admissions and unnecessary invasive testing.⁵ This clinical conundrum has led to an increasing volume of research into the role of biomarkers that aid clinicians in accurate and early diagnosis of SBIs.

Among the biomarkers that have been evaluated, C reactive protein (CRP) and procalcitonin (PCT) have consistently been reported to be reliable predictors of SBIs in paediatric patients.^{6 7} Newer studies have also proposed the emerging role of cytokines, such as interleukin-6 (IL-6), in predicting sepsis in young children.^{8 9} However, their diagnostic performances, in terms of specificity, sensitivity and areas under the receiver operating characteristic curve, have been shown to vary considerably across different paediatric populations and age groups, particularly in young children aged less than 36 months old.^{10 11} Variability between studies can be attributed in part to differences in age group, definitions of SBIs adopted by different authors and the timings of presentation of these children to the emergency departments.^{12 13}

These biomarkers have been used for different diagnostic purposes in SBI research. In using these biomarkers in clinical settings, clinicians apply different cut-off values depending on whether we are trying to rule in or rule out SBIs. Most authors choose thresholds with high sensitivity and negative predictive values in view that SBIs should not be missed.^{14 15} In a systematic review by Van den Bruel *et al*, the authors recommended PCT level of 2 ng/mL and CRP of 80 mg/L to rule in serious infection, while lower thresholds of 0.5 ng/mL for PCT and 20 mg/L for CRP were recommended to rule out SBIs.¹⁴ Importantly, the diagnostic thresholds of these biomarkers reported in various studies vary widely, limiting direct application in other populations.^{16–18} In addition, different biomarkers have different kinetic profiles in that the levels of certain biomarkers such as IL-6 rise faster than PCT and CRP, but have shorter half-lives and can become undetectable within 24 hours.¹⁹ The diagnostic performances of these biomarkers therefore may vary significantly depending on the timing of presentation of these febrile children, which may in turn be affected by availability and accessibility of health services in different countries.

In view of the evolving epidemiology of SBIs with high vaccination rates in many countries, it is also crucial to understand the performance of these biomarkers early (within 48 hours of presentation) in the course of infection.^{20 21} To date, there have been no systematic reviews or meta-analyses performed to assess the thresholds at which these biomarkers can accurately predict SBIs, nor their role in early detection of SBIs in febrile children aged ≤ 36 months old.

Objectives

The primary objective of this systematic review and meta-analysis is to determine the thresholds at which CRP, PCT

and IL-6 can perform optimally in distinguishing the presence of SBIs in children ≤ 36 months old presenting with fever without source. The secondary objective of the study is to determine their performances in early detection of SBIs within 48 hours of fever onset among children ≤ 36 months old presenting with fever without source.

METHODS AND ANALYSIS

We will report the systematic review and meta-analysis in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) Diagnostic Test Accuracy checklist.²² This review is registered with the International Prospective Register of Systematic Reviews with registration number CRD42023439093.

Eligibility criteria

We will identify, collate and review all available evidence from November 2013 to November 2023 reporting on the diagnostic accuracy of CRP, PCT and IL-6 in detecting SBIs in children aged ≤ 36 months presenting with fever without apparent source that answer our clinical question, with the following population, intervention, comparator and outcome measures. *Population*: children aged ≤ 36 months presenting with fever without source; *Intervention*: CRP, PCT and/or IL-6 performed as diagnostic tools; *Comparator*: different biomarker thresholds; *Outcomes*: detection of SBIs.

Randomised controlled trials (RCTs) and non-randomised studies including non-RCTs, controlled before-and-after studies and cohort studies will be included. Case series, case-control studies, systematic reviews, conference abstracts, studies that include both paediatric and adult patients that do not report on the paediatric population separately as well as studies not reported in English will be excluded. We will also exclude studies with small population size ($N < 50$) to prevent small-study effects that may result from reporting biases and non-generalisability,²³ as well as those in specific high-risk populations including studies that include patients with history of prematurity, immunodeficiencies and malignancies.

Information sources

We will systematically search electronic databases including MEDLINE, Cochrane Central Register of Controlled Trials, Cochrane CENTRAL, CINAHL (Cumulative Index to Nursing and Allied Health Literature) and Science Citation Index. The bibliographies of all selected studies will be reviewed and screened for eligibility to ensure that our literature search is adequate. A manual Google search will be done too.

Search strategy

In collaboration with a librarian at our institution, we have developed an electronic search strategy involving the use of controlled vocabulary and keywords related to

C-reactive protein, pro-calcitonin, interleukin-6, pediatrics, children, fever, pyrexia, infection and sepsis. Medical Subject Headings terms for MEDLINE and Cochrane will be explored as deemed appropriate. The search time frame will be from 1 July 2013 to until 31 July 2023. Duplicate results will be removed.

Study selection

The final search results will be compiled and imported into Covidence,²⁴ the software which follows the PRISMA 2009 guidelines that facilitate screening of studies during the study selection process. First, the titles and abstracts of articles will be screened and reviewed to assess their relevance. Full-text articles of all relevant articles will then be retrieved and assessed for eligibility based on inclusion and exclusion criteria. Two independent reviewers will independently screen and review the articles at both stages. Any disagreements will be resolved by a third independent reviewer or by consensus. Reasons for excluding studies will be recorded and tabulated in the appendix of the final report.

Data extraction and synthesis

Standardised data collection forms on Microsoft Excel will be used to extract relevant data from each included study independently by two researchers and then reconciled. The following data will be extracted from each article: study title, authors, publication date and year, type of publication, country, study design, sample size, inclusion and exclusion criteria, clinical setting, characteristics of study population (including age, gender, comorbidities), duration of fever at the time of diagnostic workup and method of sampling CRP, PCT and IL-6, values and units of measurement for CRP, PCT and IL-6, diagnostic performance indicators from which a 2×2 table will be tabulated (true positives (TPs), true negatives (TNs), false positives (FPs) and false negatives (FNs)), presence of documented infections, definitions of SBIs, admissions and duration of hospitalisation, as well as antibiotic use. If clarifications are needed or additional data are required, we will email the original authors. If we receive no response within 3 weeks, a reminder email will be sent. The study will be excluded if we do not hear from the authors within 1 month.

Definitions

1. Fever without source: temperature $>38^{\circ}\text{C}$ for less than 7 days without any signs or symptoms identifying source of infection.
2. Early presentation: presentation of febrile children to the emergency department within 48 hours of fever onset.²⁵
3. SBI: presence of bacteraemia, urinary tract infection (UTI), pneumonia, bacterial meningitis, osteomyelitis or septic arthritis.^{26 27}
4. Bacteraemia: growth of a single bacterial pathogen in blood, excluding growth of bacteria considered a pri-

ori as contaminants, for example, coagulase-negative *Staphylococcus*).²⁸

5. UTI: growth of single urine pathogen with at least 50 000 colony-forming units (CFUs)/mL from catheterised urine specimen; or 10 000–50 000 CFUs/mL from a catheterised specimen with positive urinalysis (positive for leucocyte esterase, nitrite or pyuria at >5 white blood cells; or at least 100 000 CFUs/mL from urine collected via voided specimens).²⁹
6. Pneumonia: lobar consolidation diagnosed on chest radiography confirmed by a paediatric radiologist.
7. Bacterial meningitis: cerebrospinal fluid leucocytes >5 cell/ μL and positive bacterial culture.
8. Osteomyelitis: positive bone scintigraphy.
9. Septic arthritis: positive bacterial culture of synovial fluid.

Risk of bias in individual studies

The validity of each study will be assessed using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS) assessment tool, which provides a structured and standardised way of evaluating the risk of bias and applicability of diagnostic test accuracy research across four domains including patient selection, index test, reference standard and flow and timing.³⁰ Any disagreements between two researchers will be resolved by a third reviewer or by consensus.

Statistical analysis

The extracted number of TPs, FPs, FN, TNs, ORs and risk ratios (RRs) from each study will be pooled to calculate sensitivity, specificity, positive likelihood ratio, negative likelihood ratio, diagnostic OR and corresponding 95% CI. The pooled diagnostic value through the summary receiver operating characteristic (SROC) curve and the area under the SROC curve will be reported too. To address our secondary objective, we will evaluate the diagnostic performances of these biomarkers in studies based on the aforementioned criteria in studies that include patients presenting early within 48 hours of fever onset. Publication bias will be investigated using funnel plots, and Egger's weighted regression method will be used to test the presence of publication bias. Secondary outcomes such as prevalence fever, pneumonia, UTI, etc will be considered as binary data and will be reported as pooled proportion with 95% CI. We will assess heterogeneity among the studies using the X^2 and I^2 tests. If $p<0.1$ or $I^2>80\%$, heterogeneity will be defined as significant and a random-effects model was used to combine the data. A fixed-effect model was chosen when $I^2\leq 50\%$ or $p\leq 0.05$. However, if there is enough number of original studies, a meta-regression will be carried out to explore the potential heterogeneity of certain index tests. In addition, subgroup analysis will be performed according to the sample size, treatment conditions and cut-off value. A sensitivity analysis will be carried out by excluding studies with high risk of bias based on QUADAS scale. If studies report ORs or RRs, then they will be assessed

using fixed-effect or random-effects model. All analyses will be carried out using SAS V.9.4, Meta-DiSc V.1.4 and R V.4.2.2 (2022-10-31 ucrt). All tests will be two sided and $p < 0.05$ will be considered as statistically significant.

DISCUSSION

While untreated bacterial infections may lead to significant morbidity and mortality, overtreatment of self-limiting viral illnesses increases the risk of antibiotic resistance and contributes to higher healthcare costs.^{1 31} Differentiating children with SBIs from those with viral illnesses is therefore essential. Indeed, multiple studies have assessed the role of biomarkers in predicting SBIs in these children.^{9 32}

CRP is an acute-phase reactant produced in the hepatocytes in response to tissue damage, infection or non-specific immune system activation which can become elevated within 24 hours of a trigger, reach its maximum within 48 hours and has a half-life of 24 hours.^{33–36} In contrast, PCT is a peptide precursor of the hormone calcitonin. Under normal physiology, only very small amount of PCT is released into circulation and its production increases rapidly within 6–12 hours and reaches its peak within 12 hours in response to bacterial endotoxins and inflammatory cytokines in extrathyroidal tissues, and has a half-life of 24 hours.³⁷ PCT has therefore been regarded as a more specific biomarker of systemic bacterial infection and sepsis, and has been reported to facilitate earlier diagnosis.^{37 38} On the other hand, IL-6 is an inflammatory cytokine produced by monocytes and macrophages which rises early in the inflammatory cascade and functions as a hepatocyte-stimulating factor that enhances the production of acute-phase reactants including CRP.^{39 40} Although IL-6 level rises sharply in response to infection, it falls rapidly with time and becomes undetectable within 24 hours.^{39 40}

While the diagnostic performances of these biomarkers have been well reported and these biomarkers have been used to fortify existing clinical decision rules, their optimal thresholds remain inconclusive due to high variability within and between studies. Additionally, their roles in early detection of SBIs for children presenting within 48 hours of fever onset are unclear. This systematic review and meta-analysis thus aims to determine the optimal thresholds at which biomarkers CRP, PCT and IL-6 can predict SBIs in young children presenting with fever without source, as well as their early detection within 48 hours of fever onset. While multiple single-centre studies have been published on the diagnostic performances of these biomarkers, this systematic review and meta-analysis will facilitate comprehensive analyses and reporting of biomarker thresholds. Our study has the potential to provide guidance on clinical decision-making in young children presenting with fever without source.

Limitations

Despite our best efforts in using a comprehensive search strategy, the keywords used may not yield all relevant

articles. In addition, we will exclude all studies reported in languages other than English, which may reduce the generalisability of the findings. Given the nature of the study, we will not have access to individual patient-level data which may render it impossible to study factors beyond those reported in published articles. In addition, there may be heterogeneity in the thresholds of the biomarkers depending on the patient population and predominant pathogen within that population.

Contributors S-LC and NS contributed to initiation and conception of the protocol. S-LC, NS, SHWY, SSMG and RS designed the protocol and data collection. S-LC, NS, SHWY, SSMG and RS drafted the manuscript and contributed to the final approval of the manuscript.

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Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval Ethical approval is not required for this study.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data sharing not applicable as no datasets generated and/or analysed for this study. This is a protocol paper for a systematic review and meta-analysis and no data have been generated.

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ORCID iDs

Natalia Sutiman <http://orcid.org/0009-0007-5713-5909>

Shu-Ling Chong <http://orcid.org/0000-0003-4647-0019>

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